1. $R \subset A \times B$

$S \subset B \times C$

$A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4, 5\}$, $C = \{1, 2, 3, 4\}$

$R = \{(1, 3), (1, 5), (2, 1), (2, 2), (3, 2), (3, 4)\}$

$S = \{(1, 2), (1, 3), (2, 2), (4, 1), (5, 4)\}$

$S \circ R = \{(1,4), (2,2), (2,3), (3,2), (3,1)\}$

$S^{-1} = \{(1,4), (2,1), (2,2), (3,1), (4,5)\}$

$R^{-1} = \{(1,2), (2,2), (2,3), (3,1), (4,3), (5,1)\}$

2. $f: \mathbb{R} \to \mathbb{R}$

$g: \mathbb{R} \to \mathbb{R}$

$f(x) = x^3$

$g(x) = \begin{cases} 3x-1, & x \geq 0 \\ -x, & x < 0 \end{cases}$

- $g \circ f = g(f(x)) = \begin{cases} 3x^3 - 1, & x^3 \geq 0 \Rightarrow 3x^3 - 1, \ x \geq 0 \\ -x, & x^3 < 0 \end{cases} = \begin{cases} 3x^3 - 1, & x \geq 0 \\ -x, & x < 0 \end{cases}$
- $f \circ f = f(f(x)) = (x^3)^3 = x^9$
- $g \circ g = g(g(x)) = \begin{cases} 3(3x-1)-1, & 3x-1 \geq 0 \\ -3x+1, & 3x-1 < 0 \end{cases} =$

$$= \begin{cases} 9x-4, & x \geq \frac{1}{3} \\ -3x+1, & x < \frac{1}{3} \end{cases}$$

- $f \circ g = f(g(x)) = \begin{cases} (3x-1)^3, & x \geq 0 \\ -x^3, & x < 0 \end{cases} =$

$$= \begin{cases} 27x^3 - 9x^2 + 3x - 1, & x \geq 0 \\ -x^3, & x < 0 \end{cases}$$